

Recorded TTC Subway Delay Patterns Across Peaks and Other Hours in Toronto, 2024*

Comparison of the Positive-delay Proportions and Median Positive-delay Durations among Three Subway Lines: Yonge-University, Bloor-Danforth and Sheppard.

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This paper examines 2024 TTC subway delay records from Toronto Open Data, comparing positive-delay proportions and median positive-delay durations across three subway lines and three time periods. The Yonge–University line had the highest proportion of positive-delay records during morning peaks, evening peaks, and other times. However, the Sheppard line had the highest median positive-delay duration during morning peaks and at other times, despite having the lowest positive-delay proportion during both peak periods. These findings show that a higher proportion of positive-delay records does not necessarily correspond to longer typical delays, highlighting the importance of considering both measures when comparing recorded subway delays.

*Code and data are available at: https://github.com/JingxuanFeng/TTC_Subway_Delay_Patterns.

1 Introduction

Subways are an important mode of transportation for commuters in Toronto. Understanding recorded subway delay patterns during peak commuting hours can help characterize service performance and identify where further investigation may be useful.

The paper uses 2024 TTC subway delay records (Toronto Transit Commission 2024b) to compare recorded delay patterns across the Yonge–University (YU), Bloor–Danforth (BD), and Sheppard (SHP) lines during morning peaks, evening peaks, and other times. We examine two aspects of those records: the proportion of records with a delay greater than zero minutes and the duration of those positive delays. A higher proportion of positive-delay records does not necessarily imply longer delays when positive durations are recorded.

Across all three time periods, Yonge–University had the highest proportion of records with positive delays. However, Sheppard had the highest median positive delay during the morning peak and at other times, despite having the lowest proportion of positive-delay records during both peak periods. These findings distinguish the prevalence of positive delays within the records from their typical duration.

The remainder of this paper is organized as follows. Section 2 describes the source and the visualizations of the data, Section 3 shows the TTC subway delay times analysis and Section 4 summarizes the results and discusses the limitations of the analysis.

2 Data

2.1 Data Source and Study Sample

We use the 2024 TTC Subway Delay Data (Toronto Transit Commission 2024b) published through Toronto Open Data (City of Toronto 2024). This dataset was selected because it records subway lines, incident times, and delay durations, allowing comparisons across lines and time periods. Our analysis focuses on the Yonge–University (line 1), Bloor–Danforth (line 2), and Sheppard (line 4) lines. The original dataset contains 26,467 records. After standardizing the YUS code to YU and retaining records coded as YU, BD, or SHP, the analysis includes 26,029 records.

The statistical programming language R (R Core Team 2026) was used to simulate data and to download, clean, test, and analyze the data. Specifically, we use `opendatatoronto` (Gelfand 2025) to download the data; `tidyverse` (Wickham et al. 2019), including `tibble` (Müller and Wickham 2026), `readr` (Wickham, Hester, et al. 2026), `dplyr` (Wickham, François, et al. 2026), `tidyr` (Wickham et al. 2025), `stringr` (Wickham 2025), and `ggplot2` (Wickham 2016), to prepare, summarize, and visualize the data; `janitor` (Firke 2024) to standardize column names; `hms` (Müller 2025b) to parse simulated times; `here` (Müller 2025a) to construct file paths; `testthat`

(Wickham 2011) to validate the cleaned data; and tinytable (Arel-Bundock 2026) to present summary tables.

2.2 Measurements and Variables

The analysis used four recorded variables: subway line, time of delay, day of the week and delay duration. Delay duration is measured in minutes using the `min_delay` field (Toronto Transit Commission 2024b). This measure describes recorded service delay rather than the additional travel time experienced by an individual passenger.

We constructed three time period categories: morning rush hour, evening rush hour, and other times. For this analysis, weekday morning rush hours are defined as 6:00–9:00 a.m. and evening rush hours as 3:00–7:00 p.m. The peak-hour boundaries follow TTC service reporting (Toronto Transit Commission 2024a). All remaining records, including weekends, are classified as other times. Public holidays are not separately excluded.

Zero minute records are retained when describing all records. These records are not treated as missing values or evidence that the subway operated on time.

2.3 Distribution of Recorded Delays

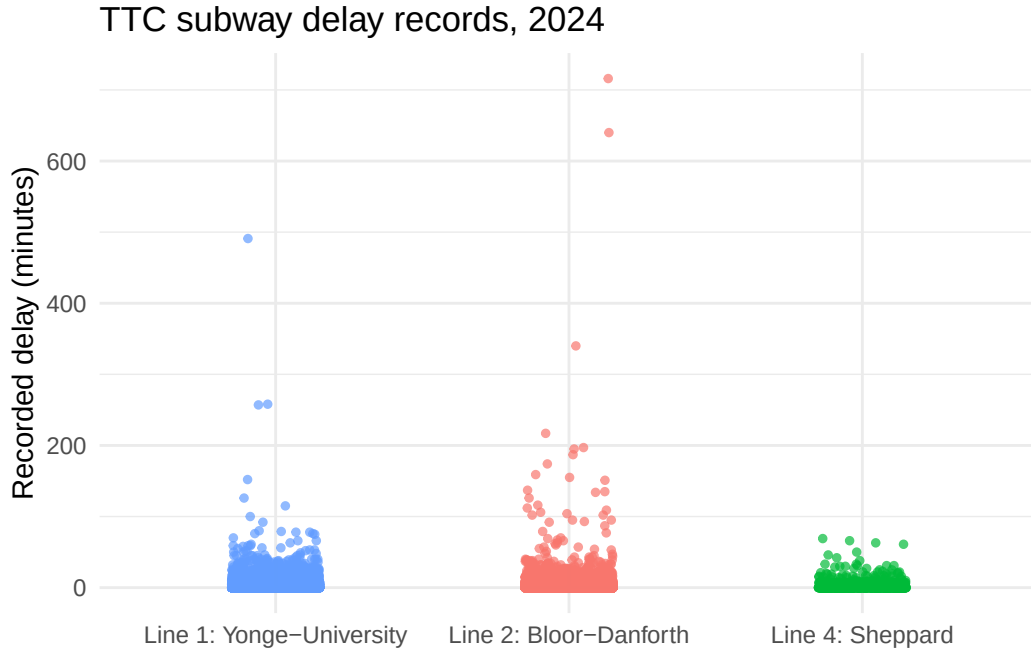


Figure 1: Recorded delay durations for all 26,029 observations, grouped by TTC subway line: Line 1 (Yonge–University), Line 2 (Bloor–Danforth), and Line 4 (Sheppard)

Figure 1 shows the recorded delay duration for all 26,029 observations. Most records have short delays, although a few show delays lasting several hours.

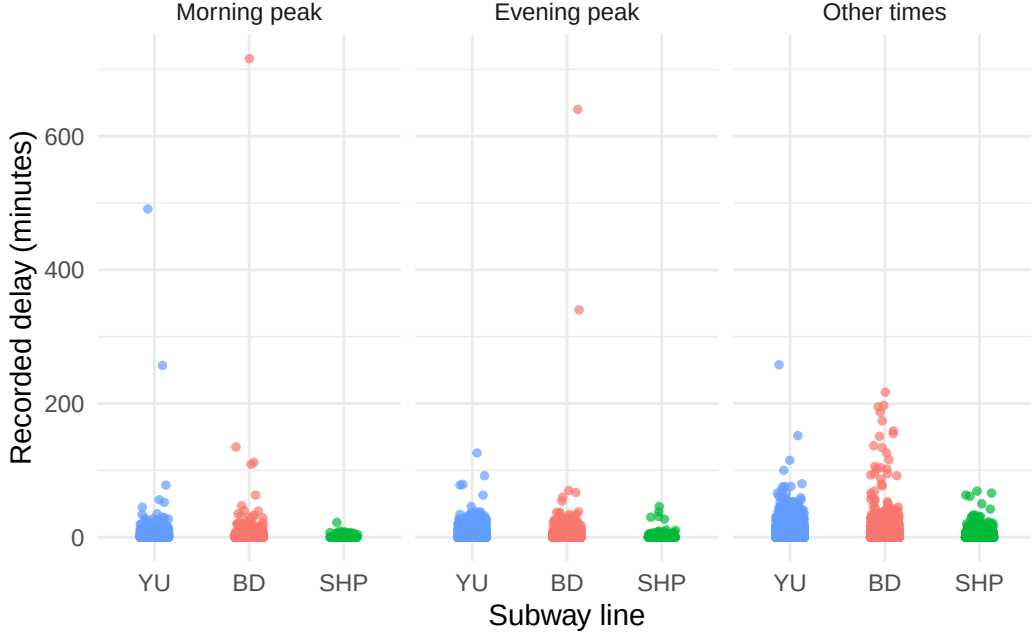


Figure 2: Recorded TTC subway delay durations (2024) by subway line: Line 1 (Yonge–University), Line 2 (Bloor–Danforth), and Line 4 (Sheppard) and time period: morning peak, evening peak and other times. Records with zero-delay included.

Figure 2 shows recorded delay durations by subway line during morning peaks, evening peaks, and other times, including records with zero delay. Across all three periods, observations are concentrated near zero, with a small number of much longer delays. The largest recorded delays occur on Bloor–Danforth line during the morning and evening peaks, while Sheppard line shows a narrower range of delay durations. However, Sheppard’s narrower range of delay durations does not establish that it performs better overall. It has far fewer records than the other two lines, and the dataset does not show how many trips or passengers each line served.

Table 1: Number of TTC subway delay records in 2024 by line: Line 1 (Yonge–University), Line 2 (Bloor–Danforth), and Line 4 (Sheppard).

Line	Number of records
Line 1: Yonge–University	14149
Line 2: Bloor–Danforth	10990
Line 4: Sheppard	890

Table 1 reports the number of records for each of the three subway lines. Both the figure and table include records with zero-minute delays.

3 Results

Table 2: Percentage of positive delay records by subway line: Line 1 (Yonge–University), Line 2 (Bloor–Danforth), and Line 4 (Sheppard) and time period: morning peak, evening peak and other times.

Line	Morning peak (%)	Evening peak (%)	Other times (%)
YU	53.1%	43.6%	37.2%
BD	37.4%	32.8%	29.8%
SHP	25.0%	27.2%	34.3%

Table 3: Median positive-delay records by subway line: Line 1 (Yonge–University), Line 2 (Bloor–Danforth), and Line 4 (Sheppard) and time period: morning peak, evening peak and other times.

Line	Morning peak delay (min)	Evening peak delay (min)	Other times delay (min)
YU	5	5	5
BD	4	4	5
SHP	6	5	7

Table 2 reports the percentage of records with positive delays for each subway line during morning peaks, evening peaks, and other times. Each percentage is calculated by dividing the number of positive-delay records by the total number of retained records within the corresponding line–period group and multiplying by 100. Table 3 reports the median delay duration in minutes for each group, using only records with positive delays. We report the median rather than the mean delay duration because a few exceptionally long delays on the Yonge–University and Bloor–Danforth lines, as shown in Figure 1, could pull the mean upward. The median better represents a typical positive recorded delay.

Table 2 shows that the Yonge–University line had the highest percentage of positive-delay records across all three periods: 53.1% during the morning peak, 43.6% during the evening peak, and 37.2% at other times. The Sheppard line had the lowest percentages during both peak periods, while the Bloor–Danforth line had the lowest percentage at other times.

Among records with positive delays, Table 3 shows that median delay durations ranged from four to seven minutes. The Yonge–University line had a median of five minutes in all three

periods. The Sheppard line had the highest median during the morning peak and at other times, at six and seven minutes, respectively. During the evening peak, its median of five minutes matched that of the Yonge–University line.

Together, the results show that a higher proportion of positive delay did not necessarily indicate longer median delays.

4 Discussion

The results highlight the distinction between the proportion of positive delays and the duration of those delays. Yonge–University had the highest proportion of positive-delay records in all three time periods, while Sheppard had the longest median positive delay during the morning peak and at other times. Considering both measures gives a fuller picture of the recorded delays across the three lines, although neither measure alone establishes overall service performance.

These comparisons have several limitations. The dataset contains incident records rather than all subway trips, so the proportion of records with positive delays cannot be interpreted as a passenger’s probability of experiencing a delay. The analysis also groups weekdays together without identifying public holidays and does not account for weather or seasonal patterns.

Future research could examine whether these patterns persist across months and investigate how delay causes differ between lines and time periods. Combining incident records with data on scheduled and actual subway operations could provide a more direct assessment of service reliability.

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